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Counterfactual Inflation Analysis: What If Ukraine Had Been Part of the Euro Area?

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1. Introduction

On 23 June 2022, the European Council granted Ukraine official candidate status for accession to the European Union. While full Euro Area membership remains a distant prospect, conditional on institutional convergence and the fulfilment of the Maastricht criteria, the question of whether Ukraine could eventually adopt the euro is no longer purely hypothetical.

This paper studies the following counterfactual question: *what would Ukraine's inflation trajectory have looked like had it been a member of the Euro Area?*

Ukraine experienced several major macroeconomic crises during the sample period: the 2008–2009 global financial crisis, the 2014–2015 Crimea annexation and Donbas conflict, and the 2022 full-scale Russian invasion. These episodes were associated with sharp hryvnia depreciations and strong inflationary pressures. The exchange rate therefore acted as a central adjustment mechanism. Under Euro Area membership, this mechanism would have been unavailable.

The analysis proceeds in two steps. First, Part A documents Ukraine's monetary regime from 2000 to 2025, identifying the periods during which the National Bank of Ukraine (NBU) exercised genuine monetary sovereignty and those during which monetary policy was already constrained by an exchange-rate anchor. This chronology is central to the identification strategy: the “treatment” of Euro Area membership is not constant over time, because Ukraine did not have the same degree of monetary autonomy across regimes.

Second, Part B constructs a counterfactual inflation path using a structural econometric framework. A preliminary common-factor exercise is used as a descriptive benchmark for Euro Area inflation dynamics. The main counterfactual is then obtained from a trivariate Structural Vector Autoregression (SVAR) with Blanchard-Quah long-run restrictions Blanchard and Quah, 1989. The model separates external oil shocks, domestic supply shocks, and demand or monetary shocks. Under hypothetical Euro Area membership, Ukraine is assumed to keep its own real and external shocks, while its demand-side monetary shocks are partly replaced by Euro Area demand shocks.

The results suggest that Euro Area membership would have reduced inflationary pressures during the main devaluation episodes, especially in 2014–2015 and 2022. However, this lower inflation would have come at the cost of losing an exchange-rate adjustment mechanism during highly asymmetric geopolitical shocks. The counterfactual therefore illustrates the classical Optimal Currency Area trade-off between credibility and flexibility.

2. Part A — Ukraine’s Monetary Regime (2000–2025)

2.1. Chronological Overview

Before constructing the counterfactual, it is necessary to understand what Ukraine actually did with its monetary sovereignty. A naive interpretation would assume that Ukraine operated under a freely floating exchange rate throughout the sample period. This is incorrect. For long periods, the NBU maintained a de facto peg to the US dollar, which strongly constrained monetary policy.

Table 1 summarises the main monetary-regime phases between 2000 and 2025. The chronology highlights that Ukraine’s monetary sovereignty was highly time-varying: limited during peg periods, more visible during inflation targeting, and severely constrained again under wartime exchange-rate controls.

Table 1: Ukraine’s Monetary Regime Chronology (2000–2025)

Period	UAH/USD	De facto regime	Capital controls		Key events	Treatment intensity
2000–Mar. 2008	$\approx 5.00\text{--}5.06$	Stabilized arrangement / de facto USD peg	Moderate		Tight exchange-rate stability. Monetary policy subordinated to the exchange-rate anchor.	Low
Apr. 2008–Feb. 2010	$4.85 \rightarrow 7.88$	Managed arrangement / crisis devaluation	Strict		Global financial crisis. Hryvnia depreciation and IMF support.	High
Mar. 2010–Jan. 2014	$\approx 7.99\text{--}8.15$	Stabilized arrangement / renewed USD peg	Moderate to strict		Re-established exchange-rate stabilization through FX intervention.	Low
Feb. 2014–Dec. 2015	$8 \rightarrow \approx 26$	Managed arrangement / crisis float	Very strict		Maidan, Crimea annexation, Donbas conflict, major hryvnia depreciation.	High
Jan. 2016–23 Feb. 2022	$\approx 25\text{--}28$	Managed float with inflation targeting	Gradually easing		Inflation targeting became the main policy framework. Policy rate used as primary instrument.	Medium–High
24 Feb. 2022–2 Oct. 2023	$29.25 \rightarrow 36.57$	Wartime peg / stabilized arrangement	Maximum		Full-scale invasion. Fixed exchange rate, strict controls, later one-off adjustment.	Medium
3 Oct. 2023–2025	$\approx 36.6\text{--}42$	Managed flexibility	Gradually easing		Partial reopening of the FX market and gradual return to flexibility.	Medium

2.2. Implications for Monetary Sovereignty

Ukraine’s monetary history reveals a fundamental asymmetry. During the peg periods of 2000–2008 and 2010–2014, monetary policy was effectively constrained by the exchange-rate anchor. With a fixed exchange rate and imperfect but meaningful capital mobility, the “impossible trinity” was binding: the NBU could not simultaneously maintain the peg, allow capital mobility, and conduct an independent monetary policy.

The crisis episodes of 2008–2009 and 2014–2015 involved sharp depreciations, but these were forced adjustments after the peg became unsustainable rather than fully discretionary policy choices. This is consistent with the “fear of floating” logic of Calvo and Reinhart (2002), where formally flexible regimes still behave as constrained exchange-rate regimes.

The clearest period of genuine monetary sovereignty is the post-2016 inflation-targeting period. From then on, the NBU increasingly relied on the policy rate as its main instrument, while allowing more exchange-rate flexibility. This is also the period during which Euro Area membership would have represented a more meaningful change in monetary regime.

The implication for the counterfactual is direct: Euro Area membership cannot be modelled as a constant treatment over the full sample. During peg periods, Ukraine had already given up much of its monetary autonomy, so the additional treatment effect of Euro Area membership should be small. During devaluation episodes and the inflation-targeting period, the effect should be larger. This time-varying treatment intensity is incorporated in the SVAR counterfactual.

3. Data

3.1. Core Inflation Data

The core dataset combines Euro Area inflation and Ukrainian inflation.

The Euro Area data consist of monthly year-on-year HICP inflation rates for 11 Euro Area countries: Austria, Belgium, Germany, Spain, Finland, France, Greece, Ireland, Italy, the Netherlands, and Portugal. These data are already expressed as annual inflation rates.

Ukraine’s CPI data are provided as a monthly price index with the previous month as base. To make Ukraine comparable with the Euro Area panel, the Ukrainian CPI series is converted into year-on-year inflation using a chained 12-month product:

$$\pi_{UKR,t}^{YoY} = \left(\prod_{k=0}^{11} \frac{P_{t-k}}{P_{t-k-1}} - 1 \right) \times 100. \quad (1)$$

This transformation is essential because the Ukrainian raw data and Euro Area data are not provided in the same format.

3.2. External Data

The SVAR framework requires macroeconomic variables beyond inflation alone. For this reason, the project augments the inflation panel with external macroeconomic series. Table 2 summarises the variables used in the empirical analysis.

Table 2: External Data Sources and Role in the Model

Variable	Source	Frequency	Role in the analysis
UAH/USD exchange rate	Stooq / external FX data	Monthly	Used to document exchange-rate regimes and major devaluation episodes.
Brent crude oil price	FRED: DCOIL-BRENTEU	Monthly	Proxy for global energy-price shocks, important given Ukraine's exposure to energy prices.
ECB deposit facility rate	FRED: ECBDFR	Monthly	Background variable for Euro Area monetary-policy conditions.
Euro Area industrial production	FRED / OECD series	Monthly	Proxy for Euro Area real activity and demand conditions.
Ukraine industry value-added growth	World Bank API	Annual, interpolated monthly	Proxy for Ukrainian real activity in the SVAR.

A limitation of the dataset is that Ukraine's real activity series is annual and must be interpolated to monthly frequency. This introduces smoothness in the output proxy and may attenuate within-year dynamics. However, the main crisis observations remain economically consistent with the chronology: the deepest contractions coincide with the 2009 global financial crisis, the 2014–2015 conflict, and the 2022 full-scale invasion.

4. Methodology

4.1. Descriptive Benchmark: Euro Area Common Inflation Factor

As a preliminary benchmark, the project extracts a common Euro Area inflation component using Principal Component Analysis, following the spirit of Ciccarelli and Mojon

(2010). The first principal component explains a large share of cross-country inflation variation, which supports the idea that Euro Area inflation contains a strong common component.

This exercise is used only as a descriptive benchmark. It is not the final counterfactual, because a simple common factor or cross-sectional Euro Area average would not provide a structural identification strategy. The final counterfactual instead relies on the SVAR shock decomposition described below.

4.2. Structural VAR with Blanchard-Quah Identification

The main empirical framework is a trivariate Structural VAR inspired by the Blanchard-Quah long-term identification approach Blanchard and Quah, 1989. The vector of variables is:

$$Y_t = \begin{pmatrix} Brent_t \\ \Delta IP_t \\ \pi_t \end{pmatrix}, \quad (2)$$

where $Brent_t$ is the Brent oil price, ΔIP_t is industrial production growth or real activity growth, and π_t is year-on-year inflation.

The inclusion of Brent extends the standard output-inflation framework by explicitly capturing global energy shocks. This is important for Ukraine because energy-price and external supply shocks are central to its inflation dynamics.

The reduced-form VAR is:

$$Y_t = c + \sum_{j=1}^p A_j Y_{t-j} + u_t, \quad (3)$$

where u_t are reduced-form residuals and p is the lag order. In the implementation, the lag order is selected by information criteria and capped to preserve degrees of freedom.

4.3. Identification Restrictions

The Blanchard-Quah identification relies on long-run restrictions. In the standard output-inflation framework, demand shocks are assumed to have no permanent effect on output. In this trivariate extension, the structural shocks are interpreted as:

- **Oil shocks:** external global energy-price shocks;
- **Supply shocks:** real domestic shocks affecting output and inflation;
- **Demand or monetary shocks:** aggregate demand and monetary-policy shocks.

Let the long-run multiplier matrix be:

$$C(1) = \left(I - \sum_{j=1}^p A_j \right)^{-1}. \quad (4)$$

The structural impact matrix is obtained using the long-run covariance matrix:

$$B_0 = C(1)^{-1} \cdot \text{chol}(C(1)\Sigma_u C(1)'), \quad (5)$$

where Σ_u is the covariance matrix of reduced-form residuals. This decomposition allows us to recover structural shocks from the reduced-form VAR residuals.

4.4. Impulse Response Function Validation

Figure 1 presents the cumulative impulse response functions from the Blanchard-Quah SVAR estimation. The key validation check is that demand shocks have a near-zero cumulative effect on industrial production at long horizons for both Ukraine and the Euro Area. This confirms that the long-run identification restriction is empirically satisfied.

The IRFs also illustrate the structural difference between the two economies: Ukraine's inflation response to both supply and demand shocks is larger in magnitude and more persistent than the Euro Area's, reflecting the lower credibility of the NBU's monetary framework and the greater exposure to asymmetric real shocks.

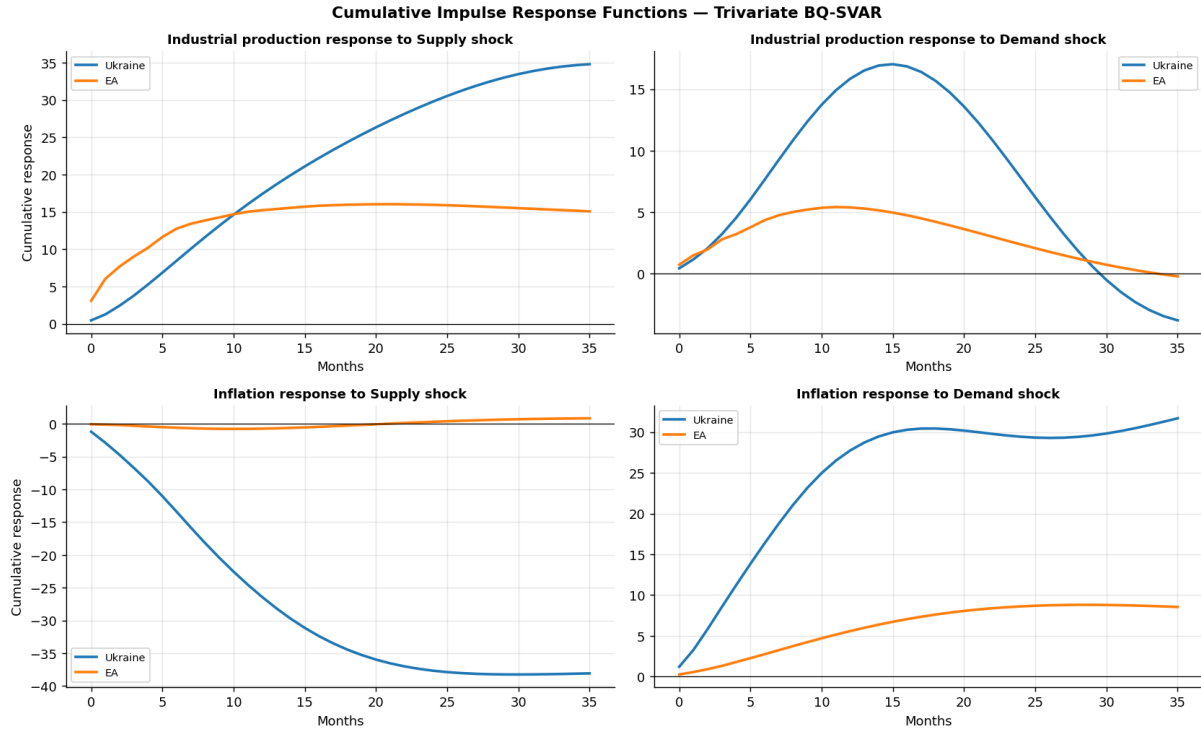


Figure 1: Cumulative impulse response functions — Blanchard-Quah trivariate SVAR. Ukraine (blue) and Euro Area (orange). The demand shock converges to zero effect on industrial production at the 36-month horizon, validating the long-run identification restriction.

4.5. Counterfactual Construction

The counterfactual assumption is the following:

Under Euro Area membership, Ukraine would have kept its own external and supply shocks, but its demand or monetary shocks would have been partly replaced by Euro Area demand shocks, reflecting ECB monetary-policy conditions.

This is consistent with the economic meaning of Euro Area membership: a country loses its national monetary policy but remains exposed to domestic real shocks and external shocks.

The counterfactual replacement is not applied uniformly across all periods. Instead, the shock replacement is weighted by the monetary-regime chronology from Part A. Let w_t denote the treatment intensity weight. Low values of w_t correspond to periods where Ukraine was already constrained by a peg. Higher values correspond to crisis or floating periods in which Euro Area membership would have represented a stronger change in monetary regime.

The counterfactual inflation residual is constructed as:

$$\varepsilon_{\pi,t}^{CF} = w_t \cdot \varepsilon_{demand,t}^{EA} \cdot \sigma + (1 - w_t) \cdot \varepsilon_{\pi,t}^{UKR}, \quad (6)$$

where

$$\sigma = \frac{sd(\varepsilon_{\pi}^{UKR})}{sd(\varepsilon_{demand}^{EA})} \quad (7)$$

rescales Euro Area demand shocks to Ukraine’s inflation-residual volatility.

The final counterfactual path is then obtained by recursively simulating Ukraine’s VAR:

$$\hat{Y}_t^{CF} = \hat{c} + \sum_{j=1}^p \hat{A}_j \hat{Y}_{t-j}^{CF} + \varepsilon_t^{CF}. \quad (8)$$

Table 3: Treatment Intensity Weights by Monetary Regime

Period	Regime	Treatment intensity w_t
2006–2008	De facto USD peg	Low
2008–2009	GFC devaluation	High
2010–2014	Renewed USD peg	Low
2014–2015	Crimea / Donbas crisis	High
2016–2022	Inflation targeting	Medium–High
2022–2023	Wartime peg and controls	Medium
2023–2025	Managed flexibility	Medium

The exact numerical weights are calibrated in the notebook and documented in the code. They should be interpreted as a regime-consistent treatment intensity rather than as estimated structural parameters.

5. Results

5.1. Main Counterfactual Inflation Path

Figure 2 presents the main result of the paper. It compares Ukraine’s actual year-on-year inflation with the SVAR counterfactual inflation path under hypothetical Euro Area membership.

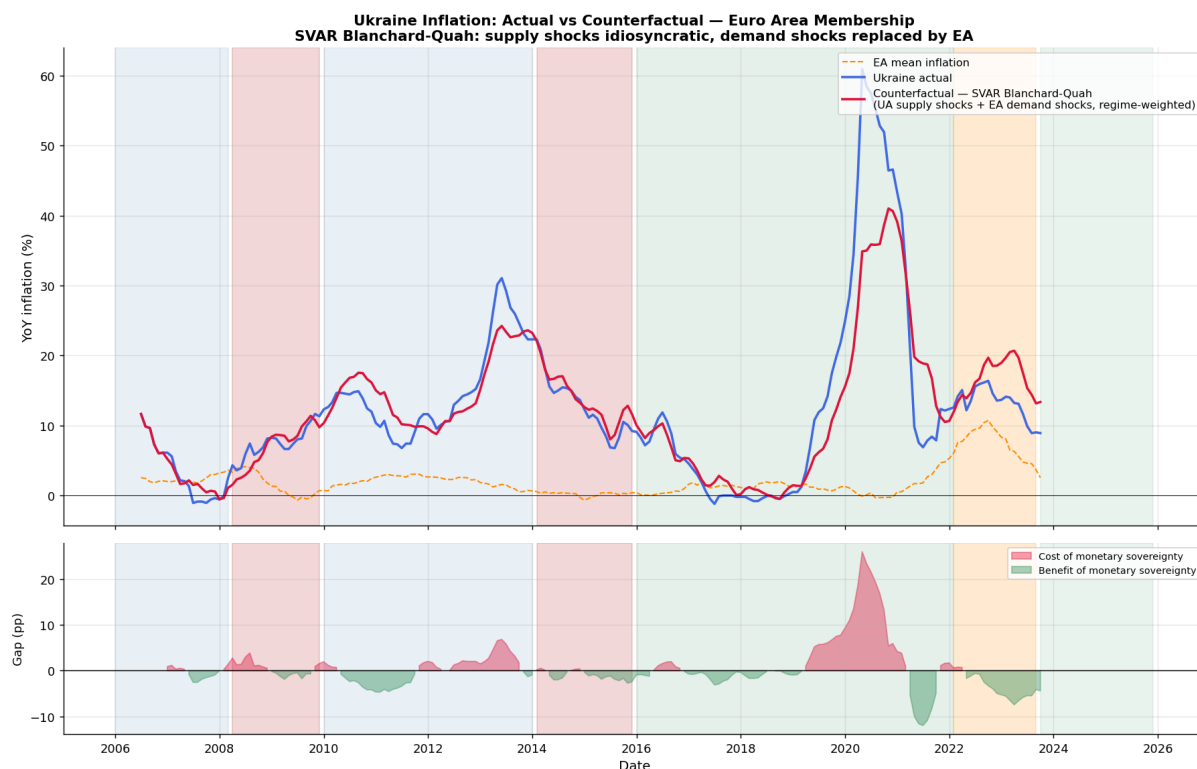


Figure 2: Ukraine inflation: actual vs counterfactual under hypothetical Euro Area membership. The counterfactual is constructed using a trivariate SVAR with Blanchard-Quah identification and regime-dependent treatment intensity.

The figure shows that the counterfactual path remains close to actual inflation during periods where Ukraine was already constrained by an exchange-rate peg. This is consistent with the Part A regime analysis: when monetary policy was already subordinated to the exchange-rate anchor, Euro Area membership would not have represented a major additional loss of monetary autonomy.

The gap becomes more visible during major crisis episodes, especially in 2014–2015 and 2022. These are precisely the periods in which exchange-rate depreciation generated strong inflationary pressures. Under Euro Area membership, the exchange-rate pass-through channel would have been absent, and inflation would likely have been more anchored.

5.2. Average Results by Regime

Table 4 summarises the average actual and counterfactual inflation rates by broad monetary-regime period. The values are approximate period averages and should be interpreted as descriptive summaries of the model output rather than as exact point estimates.

Table 4: Average Counterfactual Results by Monetary-Regime Period

Period	Avg. actual inflation	Avg. counterfactual	Avg. gap
USD peg I (2006–2008)	$\approx 10\%$	$\approx 11\%$	≈ 0 pp
GFC 2008–2009	$\approx 7\%$	$\approx 9\%$	small
USD peg II (2010–2013)	$\approx 10\%$	$\approx 11\%$	≈ 0 pp
Crimea/Donbas 2014–2015	$\approx 26\%$	$\approx 20\%$	+6 pp
Inflation targeting 2016–2021	$\approx 9\%$	$\approx 10\%$	small
Full-scale invasion 2022	$\approx 28\%$	$\approx 20\%$	+8 pp

The key result is not that Euro Area membership would always have produced lower inflation. Rather, the effect depends on the monetary regime and on the nature of the shock. The counterfactual gap is small during peg periods, but larger during crises where domestic currency depreciation amplified inflation.

6. Economic Interpretation

The counterfactual implies that Ukraine’s monetary sovereignty was a double-edged instrument whose inflationary cost and macroeconomic benefit varied sharply across regimes.

6.1. The 2008–2009 Global Financial Crisis

During the 2008–2009 crisis, the counterfactual remains relatively close to actual inflation. This reflects the fact that Ukraine had operated under a de facto USD peg before the crisis. Monetary policy was already constrained, and the depreciation that occurred in 2008–2009 was largely a forced adjustment after the peg became unsustainable.

Under Euro Area membership, Ukraine would not have experienced the same nominal exchange-rate depreciation against the euro. However, the adjustment would likely have occurred through internal devaluation, with pressure on wages, prices, and output. Therefore, the inflation benefit of Euro Area membership in this episode is not mechanically large.

6.2. The 2014–2015 Crimea and Donbas Crisis

The 2014–2015 episode is the clearest case where monetary sovereignty was associated with large inflationary adjustment costs. The collapse of the hryvnia strongly increased import prices and destabilised inflation expectations. In the counterfactual path, inflation is lower and smoother, suggesting that Euro Area membership would have reduced the exchange-rate pass-through component of inflation.

However, this does not mean that Euro Area membership would have eliminated the crisis. The shock was highly asymmetric and geopolitical. Without its own currency and central bank liquidity backstop, Ukraine could have faced a different type of crisis, potentially through sovereign-risk and fiscal-stress channels, as emphasised by De Grauwe (2012).

6.3. The 2022 Full-Scale Invasion

The 2022 invasion is more ambiguous. The counterfactual suggests that Euro Area membership would have limited part of the inflationary pressure associated with exchange-rate adjustment. The absence of a domestic currency depreciation would likely have reduced import-price inflation.

At the same time, the shock was extremely asymmetric. The ECB's monetary policy was calibrated for the Euro Area aggregate, not for a wartime Ukrainian economy. A single monetary policy could therefore have been inappropriate for Ukraine's specific conditions. This is exactly the type of concern highlighted by Optimal Currency Area theory: common monetary policy is costly when shocks are asymmetric and adjustment mechanisms are limited (Mundell, 1961; Bayoumi and Eichengreen, 1993).

6.4. Overall Assessment

Overall, the results support a nuanced interpretation. Euro Area membership would likely have imported credibility and reduced inflation volatility during major devaluation episodes. This is consistent with the credibility-import channel discussed by Giavazzi and Pagano (1988).

However, monetary sovereignty also provided Ukraine with an adjustment mechanism during large asymmetric shocks. The exchange rate absorbed part of the real shock, but at the cost of higher inflation. The counterfactual therefore does not imply that Euro Area membership would have been unambiguously better. It shows a trade-off: lower inflation and greater nominal credibility versus less flexibility in the face of country-specific shocks.

7. Limitations and Conclusion

7.1. Limitations

Several limitations should be noted.

First, the Ukrainian real-activity variable is interpolated from annual data to monthly frequency. This is necessary because consistent monthly Ukrainian output data are difficult to obtain for the full sample, but it smooths short-run dynamics and may affect the SVAR estimation.

Second, the treatment intensity weights are calibrated from the Part A monetary-regime chronology rather than estimated endogenously. A more structural approach could model regime-switching explicitly.

Third, the counterfactual focuses on inflation dynamics. It does not fully model other macroeconomic outcomes that would have been central under Euro Area membership, such as sovereign spreads, fiscal stress, unemployment, or output losses.

Fourth, the analysis assumes that Euro Area membership mainly changes demand or monetary shocks. In reality, membership could also have affected institutions, financial integration, expectations, and capital flows over the long run.

7.2. Conclusion

This paper constructs a counterfactual inflation path for Ukraine under hypothetical Euro Area membership using a regime-weighted SVAR with Blanchard-Quah identification.

The first contribution is to show that Ukraine’s monetary regime was not constant over the sample. During peg periods, Ukraine already had limited monetary autonomy, so the additional effect of Euro Area membership would likely have been small. During devaluation episodes and the inflation-targeting period, monetary sovereignty mattered more.

The second contribution is to construct a structural counterfactual that goes beyond a mechanical Euro Area average. By replacing Ukraine-specific demand or monetary shocks with Euro Area demand shocks while preserving external and supply shocks, the model provides an economically interpretable counterfactual inflation path.

The results suggest that Euro Area membership would likely have reduced inflationary pressures during the 2014–2015 and 2022 crises by removing the domestic exchange-rate depreciation channel. However, this would have come at the cost of losing monetary and exchange-rate flexibility during highly asymmetric shocks.

The conclusion is therefore not that Euro Area membership would have been clearly superior or inferior. Rather, the counterfactual illustrates the central Optimal Currency Area trade-off: credibility and inflation anchoring on one side, flexibility and shock absorption on the other.

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